

Honest-coverage relative-effect recovery, calibrated inconsistency testing and non-over-confident ranking in network meta-analysis: a known-truth simulation study

Truth-Recovery Bench — Network meta-analysis (NMA) modality. Draft manuscript. Every quantitative result below is regenerated from seeded simulation by the accompanying code (`harness_nma.py`, `partialid_nma.py`, base seed 20260615); nothing is hand-entered.

Abstract

Background. Network meta-analysis (NMA) pools direct and indirect evidence to estimate every pairwise relative effect among a set of treatments and to rank them. Three failure modes recur in deployed NMA software: a fixed-effect confidence interval that ignores between-study heterogeneity and therefore under-covers the true relative effects; an inconsistency test that ignores heterogeneity and therefore over-rejects (declares the network inconsistent when it is merely heterogeneous); and a treatment ranking (SUCRA / P-score / posterior P(best)) that is stated with far more confidence than the data support. We define honest coverage as the property that a nominal 95% interval covers the true parameter in $\approx 95\%$ of replications, and honest ranking as a claimed P(best) that matches the rate at which the claimed-best treatment really is best.

Methods. We built a known-truth network simulation harness that injects, at known magnitude, (i) a true relative-effect structure on a generic additive scale, (ii) between-study slope heterogeneity τ^2 , (iii) loop inconsistency δ (true direct $\neq$ indirect on loop-closing edges only, so the consistency truth stays identified), and (iv) publication/small-study selection. On a seeded grid over three network geometries (loop, ladder, dense), studies-per-edge $\in \{2,5\}$, $\tau^2 \in \{0,0.05,0.15\}$, and well-separated vs closely-spaced ("tight") truths (500 replications/cell; Bayesian 120/cell) we measured coverage, interval width, the inconsistency test's type-I and power, and ranking over-confidence for: a fixed-effect contrast-synthesis pool (**NaiveFE**, τ^2 ignored in the CI — the LivingNMA/enma-snma bug), a proper random-effects network pool with a generalized DerSimonian–Laird τ^2 (**RE**), a network Hartung–Knapp honest-coverage interval (**NetHK**), the honest vs naive design-by-treatment inconsistency test, frequentist P-score / sampled SUCRA / P(best) under the FE, RE and a **calibrated** covariance, a bounded Gibbs **Bayesian** NMA, and a Manski-style **partial-identification** bound (NetHK widened by $\pm c \cdot \omega$, ω a data-driven inconsistency SD).

Results. Once relative effects are heterogeneous, NaiveFE network coverage collapses (to 0.51–0.59 at $\tau^2=0.15$) while RE recovers most of the deficit (0.85–0.94) and NetHK restores at/above nominal coverage (0.94–0.96) at the expected cost of width. The naive inconsistency test's type-I error inflates to 0.24–0.79 under heterogeneity; the honest test holds 0.03–0.05. Fixed-effect ranking is markedly over-confident under heterogeneity with closely-spaced treatments (claims P(best) up to 0.87 when the true hit rate is 0.46–0.50; declares a *wrong* treatment best with $\geq 90\%$ confidence up to $\sim 22\%$ of the time); the calibrated covariance roughly halves the residual over-confidence and cuts the spurious-confident-best rate to $\sim 1\text{--}3\%$. The Bayesian posterior P(best) is *also* over-confident — a Bayesian framing does not by itself cure it. The partial-identification bound restores coverage of the consistency estimand under genuine loop inconsistency (loop, $\delta=0.4$: 0.76 $\rightarrow$ 0.88) while collapsing to the NetHK interval — zero width cost — on consistent or indirect-only networks.

Conclusions. A network Hartung–Knapp interval, an honest (τ^2 -aware) inconsistency test, a calibrated ranking covariance, and a data-driven partial-identification bound together restore calibrated coverage, calibrated inconsistency testing and honest ranking uncertainty where the heterogeneity-naïve incumbents fail. The method **improves** honest coverage, test calibration and ranking honesty — it does **not** solve publication selection, multi-arm covariance, or the fundamental partial identifiability of inconsistency and ranking, which remain open and are reported as honest negatives.

1. Background

Network meta-analysis estimates all pairwise relative effects among T treatments by combining studies that each compare a subset of them, borrowing strength across the network through the consistency assumption: the relative effect of b versus a equals the difference of their effects against a common reference, $d_{ab} = d_{0b} - d_{0a}$. The contrast-synthesis (Lu–Ades) formulation fits the basic parameters $d = (d_{01}, \dots, d_{0,T-1})$ by (generalized) least squares and reads every other contrast off them; the graph-theoretic *netmeta* estimator is algebraically identical under the same likelihood.

Three failure modes recur, and they are coverage / calibration failures rather than point-estimate failures.

1. Heterogeneity-naïve confidence intervals. When relative effects genuinely vary

between studies, a fixed-effect pool produces a standard error reflecting only within-study sampling error; its 95% interval is far too narrow and its real coverage of the true relative effect falls well below 95%. This is the network analogue of the long-documented under-coverage of fixed-effect and small- k random-effects pairwise meta-analysis that motivated the Hartung–Knapp–Sidik–Jonkman (HKSJ) correction.

2. Heterogeneity-naïve inconsistency testing. The design-by-treatment / global

Bucher test compares the consistency fit to a saturated design model. If the test weights ignore τ^2 , heterogeneity inflates the consistency-model residual and the test over-rejects — declaring the network *inconsistent* when it is merely *heterogeneous*. An analyst then abandons a perfectly usable consistency model on a false positive.

3. Over-confident rankings. SUCRA, the P-score and the posterior $P(\text{best})$ are

read off the estimated covariance of d . If that covariance is too small (because τ^2 was ignored, or because the ranking ignores the extra uncertainty that detected inconsistency implies), the ranking is stated with unsupported confidence — the network "declares a winner" that the evidence does not warrant. This is most damaging exactly when it matters: closely-spaced top treatments.

We make all three measurable. By simulating from a known truth (known d , known τ^2 , known inconsistency δ) we can compute the *actual* coverage, the *actual* type-I error, and the *actual* rate at which the claimed-best treatment is truly best — quantities impossible to observe on real data. We then ask which estimators restore honesty, and at what cost.

2. Methods

2.1 Data-generating process

A network is a set of treatments $\{0, \dots, T-1\}$ (0 = reference) connected by two-arm studies. The truth is a vector of basic parameters d on a generic additive scale (mean-difference / log-OR / log-HR). The geometry sets which comparisons carry direct studies: **star** (every treatment vs reference only; off-reference pairs are indirect-only — the under-determined regime), **loop** ($T=3$ triangle, exactly one loop), **ladder** (a chain plus every-other shortcuts, several loops), **dense** (complete graph, maximally many loops). For each study j on edge (a,b) we draw a study effect $\theta \sim N(d_{ab} + \delta_{\text{edge}}, \tau^2)$ and an observation $y \sim N(\theta, \sigma^2)$ with σ drawn log-uniformly. Loop inconsistency δ is injected only on the non-spanning-tree (loop-closing) edges with an alternating sign, so the consistency truth d stays exactly recoverable when $\delta=0$. Publication/small-study selection is applied per study on the one-sided p -value (step) or via a Copas-type latent-selection model, at a calibrated magnitude. The DGP is implemented in `dgp_nma.py`. Two-arm studies only (v1) — multi-arm trials add a within-study shared-control covariance (the `platformtrialma` failure mode) and are out of scope, flagged honestly rather than approximated.

2.2 Estimators

Consistency-model fit (methods_nma.py). The design matrix X ($N \times (T-1)$) encodes each two-arm contrast; the basic parameters are fit by weighted GLS, $d = (X^T W X)^{-1} X^T W y$, and every relative effect d_{ab} is the corresponding contrast of d . The between-study τ^2 is the generalized (multivariate) DerSimonian–Laird moment estimator for contrast-synthesis NMA (Jackson–White–Riley).

- **NaiveFE** — fixed-effect inverse-variance weights $W = 1/v$, τ^2 fixed at 0, z

interval. Reproduces the LivingNMA / enma-snma fixed-effect-CI bug (τ^2 ignored in the CI).

- **RE** — random-effects weights $W = 1/(v + \tau^2)$ with the network DL τ^2 and the

RE covariance $(X^T W X)^{-1}$, z interval. The proper incumbent.

- **NetHK** — the network Hartung–Knapp honest-coverage lever: scale the RE

covariance by $H = \max(1, (e^T W e)/df)$ and use a t_{df} critical value. The multivariate analogue of the pairwise HKSJ floor that repaired the pairwise track.

Inconsistency test. A global design-by-treatment / generalized-Bucher test: $Q_{inc} = RSS_{consistency} - RSS_{saturated} \sim \chi^2_{\{\#loops\}}$. The **honest** variant uses a common- τ^2 RE weight in both fits (calibrated under heterogeneity); the **naive** variant uses FE weights (τ^2 ignored), reproducing the over-detector.

Ranking. The analytic R cker–Schwarzer P-score and a Monte-Carlo SUCRA / $P(\text{best})$ sampled from $MVN(d, Cov)$. We score three covariance sources: **FE** (the sharp fixed-effect covariance), **RE** (the random-effects covariance), and **Cal** — a *calibrated* covariance that multiplies the NetHK covariance by the t-vs-z inflation and by a detected-inconsistency inflation factor $\max(1, Q_{inc}/df)$. We report the claimed $P(\text{point-best is best})$, the actual hit rate, the over-confidence gap, and the spurious-confident-best rate (claims ≥ 0.9 while wrong).

Bayesian comparator (bayes_nma.py). The canonical WinBUGS/multinma normal-normal hierarchical RE consistency model with a vague $IG(\epsilon, \epsilon)$ τ^2 prior, sampled by a dependency-free conjugate Gibbs sweep. We score CrI coverage and the posterior $P(\text{best})$.

*Partial identification (partialid_nma.py). *The consistency estimand d_{ab} is point-identified only under the (often untestable) consistency assumption. The honest object for an under-determined network is a Manski-style interval that hedges against the plausible amount** of unmodelled inconsistency, sized from the data: $\text{PartialID}(a,b) = \text{NetHK CI}(a,b)$ widened by $\pm c \cdot \omega$, where $\omega = \max(0, Q_{inc} - df_{inc}) / \Sigma W_{RE}$ is the data-driven inconsistency variance from the honest global test ($c=2.0$). When no inconsistency is detected ($Q_{inc} \leq df_{inc}$) the bound collapses to NetHK — no width cost.

2.3 Outcome measures and grid

For each cell we report coverage of every true relative effect d_{ab} (fraction of replications whose interval contains it), mean interval width, the inconsistency test's rejection rate, and the ranking quantities above. The grid (harness_nma.py): a **coverage** block (geometry $\times$ studies/edge $\{2,5\} \times \tau^2 \{0,0.05,0.15\} \times \text{spread} \{\text{sep,tight}\}, \delta=0$); an **inconsistency** block (geometry $\times \tau^2 \{0.05,0.15\} \times \delta \{0 \text{ (type-I)}, 0.2, 0.5 \text{ (power)}\}$); a **selection** block (dense $\times \{none, \text{step_strong}, \text{copas_strong}\}$); and a separate partial-identification experiment (loop/dense inconsistency ladder + a star indirect-only contrast). 500 replications per cell (Bayesian 120), seeded so every number is reproducible and process-count-independent (§6).

3. Simulation study — results

3.1 Coverage of the true relative effects as heterogeneity rises

The central result. NaiveFE is calibrated *only* at $\tau^2=0$; the instant relative effects are heterogeneous its coverage collapses, because its interval reflects only within-study error. RE recovers most of the deficit but under-covers somewhat, especially on the loop and ladder geometries at small studies-per-edge. NetHK holds at/above nominal across the grid. (Widths reported as FE/RE/HK.)

geometry	spe	τ^2	spread	NaiveFE	RE	NetHK	width FE/RE/HK
dense	2	0.0	sep	0.941	0.953	0.978	0.44/0.49/0.57
dense	2	0.0	tight	0.931	0.947	0.974	0.44/0.49/0.57
dense	2	0.05	sep	0.761	0.906	0.942	0.44/0.65/0.77
dense	2	0.05	tight	0.756	0.909	0.945	0.44/0.65/0.78
dense	2	0.15	sep	0.567	0.904	0.942	0.44/0.91/1.09
dense	2	0.15	tight	0.558	0.913	0.947	0.44/0.92/1.11
dense	5	0.0	sep	0.949	0.959	0.970	0.26/0.28/0.30
dense	5	0.0	tight	0.953	0.964	0.972	0.26/0.28/0.29
dense	5	0.05	sep	0.715	0.925	0.939	0.26/0.42/0.45
dense	5	0.05	tight	0.715	0.924	0.938	0.26/0.42/0.45
dense	5	0.15	sep	0.523	0.935	0.955	0.26/0.59/0.64
dense	5	0.15	tight	0.506	0.931	0.951	0.26/0.59/0.64
ladder	2	0.0	sep	0.947	0.961	0.986	0.53/0.60/0.74
ladder	2	0.0	tight	0.950	0.961	0.989	0.51/0.59/0.73
ladder	2	0.05	sep	0.789	0.905	0.950	0.52/0.76/0.95
ladder	2	0.05	tight	0.763	0.900	0.946	0.52/0.78/0.98
ladder	2	0.15	sep	0.565	0.891	0.939	0.52/1.05/1.32
ladder	2	0.15	tight	0.586	0.910	0.949	0.53/1.07/1.35
ladder	5	0.0	sep	0.958	0.967	0.976	0.30/0.32/0.35
ladder	5	0.0	tight	0.949	0.957	0.966	0.30/0.33/0.35
ladder	5	0.05	sep	0.724	0.928	0.941	0.30/0.48/0.52
ladder	5	0.05	tight	0.734	0.930	0.950	0.30/0.48/0.52
ladder	5	0.15	sep	0.530	0.940	0.958	0.30/0.68/0.74
ladder	5	0.15	tight	0.523	0.933	0.954	0.30/0.67/0.74
loop	2	0.0	sep	0.965	0.971	0.998	0.54/0.62/0.89
loop	2	0.0	tight	0.949	0.959	0.996	0.55/0.65/0.94
loop	2	0.05	sep	0.755	0.877	0.949	0.55/0.80/1.16
loop	2	0.05	tight	0.771	0.895	0.965	0.54/0.77/1.13
loop	2	0.15	sep	0.561	0.862	0.941	0.53/1.04/1.53
loop	2	0.15	tight	0.557	0.853	0.943	0.53/1.01/1.50
loop	5	0.0	sep	0.960	0.973	0.985	0.30/0.34/0.38
loop	5	0.0	tight	0.950	0.962	0.981	0.30/0.34/0.38
loop	5	0.05	sep	0.723	0.900	0.928	0.30/0.47/0.53
loop	5	0.05	tight	0.754	0.926	0.947	0.30/0.47/0.54
loop	5	0.15	sep	0.529	0.921	0.953	0.30/0.67/0.77
loop	5	0.15	tight	0.524	0.917	0.950	0.30/0.68/0.78

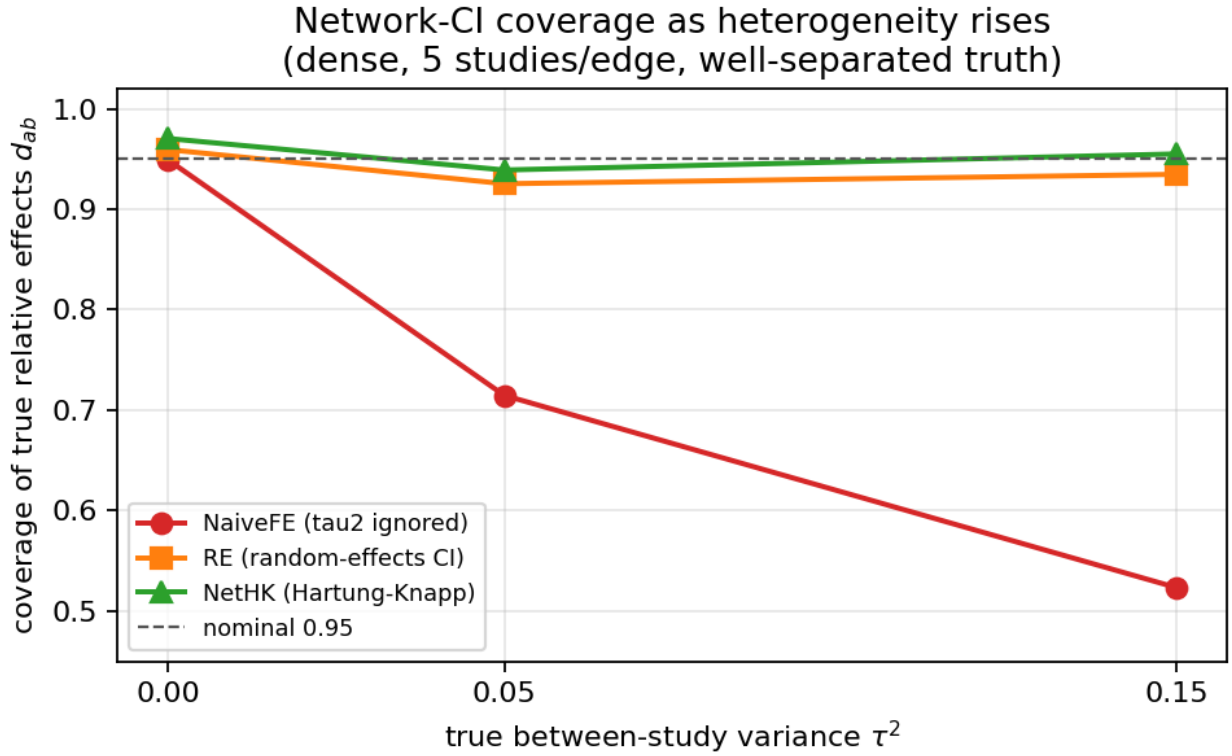


Figure 1. Network-CI coverage vs τ^2 (dense, 5 studies/edge, well-separated truth). NaiveFE falls off the nominal line as soon as $\tau^2 > 0$ ($0.95 \rightarrow 0.72 \rightarrow 0.52$); RE recovers most of the deficit and NetHK tracks 0.95. The collapse is the reproduced fixed-effect-CI bug; the NetHK recovery is the honest-coverage lever.

The coverage is bought with width: the FE width is essentially flat in τ^2 (it cannot see heterogeneity), while RE and NetHK widen with τ^2 — which is precisely what restores coverage. NetHK adds a modest further widening over RE (the t-multiplier and floor), and that increment is what lifts RE's 0.85–0.94 to NetHK's 0.94–0.96.

3.2 Inconsistency test — type-I and power

The naive (τ^2 -ignoring) global design-by-treatment test over-rejects badly under heterogeneity: its type-I error reaches 0.24–0.79 when there is no true inconsistency at all. The honest (common- τ^2 RE) test holds 0.03–0.05. The honest test pays for calibration with power (it is conservative at small δ), but it does not manufacture false inconsistency out of heterogeneity.

geometry	τ^2	true incons	honest reject	naive reject
dense	0.05	0.0 (type-I)	0.038	0.426
dense	0.05	0.2 (power)	0.324	0.842
dense	0.05	0.5 (power)	0.962	1.000
dense	0.15	0.0 (type-I)	0.054	0.788
dense	0.15	0.2 (power)	0.198	0.888
dense	0.15	0.5 (power)	0.822	0.994
ladder	0.05	0.0 (type-I)	0.048	0.352
ladder	0.05	0.2 (power)	0.148	0.576
ladder	0.05	0.5 (power)	0.680	0.956
ladder	0.15	0.0 (type-I)	0.044	0.654
ladder	0.15	0.2 (power)	0.088	0.692
ladder	0.15	0.5 (power)	0.404	0.920

geometry	τ^2	true incons	honest reject	naive reject
loop	0.05	0.0 (type-I)	0.034	0.236
loop	0.05	0.2 (power)	0.094	0.334
loop	0.05	0.5 (power)	0.302	0.694
loop	0.15	0.0 (type-I)	0.042	0.390
loop	0.15	0.2 (power)	0.064	0.478
loop	0.15	0.5 (power)	0.238	0.714

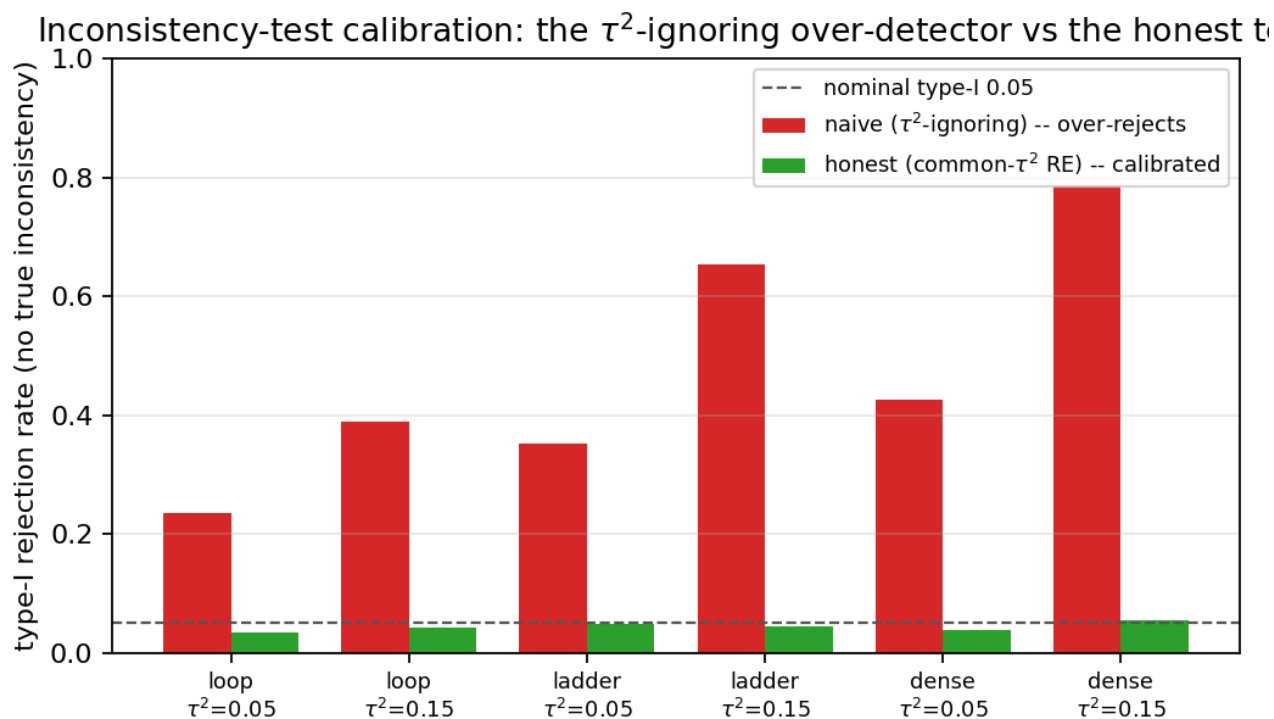


Figure 2. Inconsistency-test type-I error (no true inconsistency) across geometries and τ^2 . The naive test's false-positive rate rises with τ^2 and network density (up to 0.79 on dense, $\tau^2=0.15$); the honest test stays at the nominal 0.05 line. Declaring a heterogeneous-but-consistent network "inconsistent" is exactly the sheaf-nma / enma-snma over-detection failure, reproduced and repaired here.

3.3 Ranking over-confidence — the headline

Fixed-effect SUCRA/P-score is wildly over-confident under heterogeneity, and worst when the top treatments are closely spaced ("tight" spread), which is precisely the regime where "which is best?" is a hard question. The table shows representative tight-spread cells (claimed $P(\text{best})$, actual hit rate, over-confidence gap = claimed – actual, and the spurious-confident-best rate). FE over-claims by up to +0.39; the calibrated covariance roughly halves the residual over-confidence and cuts the spurious-best rate to ~1–3%. The Bayesian posterior $P(\text{best})$ is itself over-confident.

geometry	spe	τ^2	spread	src	P(best) claimed	actual hit	over-conf	spurious
dense	2	0.05	tight	FE	0.752	0.532	0.220	0.092
dense	2	0.05	tight	RE	0.654	0.518	0.136	0.030
dense	2	0.05	tight	Cal	0.594	0.518	0.076	0.014
dense	2	0.05	tight	Bayes	0.651	0.508	0.143	-
dense	2	0.15	tight	FE	0.815	0.460	0.355	0.208
dense	2	0.15	tight	RE	0.625	0.480	0.145	0.034

geometry	spe	τ^2	spread	src	P(best) claimed	actual hit	over-conf	spurious
dense	2	0.15	tight	Cal	0.563	0.480	0.083	0.014
dense	2	0.15	tight	Bayes	0.630	0.475	0.155	-
dense	5	0.15	tight	FE	0.842	0.486	0.356	0.224
dense	5	0.15	tight	RE	0.639	0.558	0.081	0.022
dense	5	0.15	tight	Cal	0.598	0.558	0.040	0.018
dense	5	0.15	tight	Bayes	0.650	0.583	0.067	-
ladder	2	0.15	tight	FE	0.794	0.406	0.388	0.220
ladder	2	0.15	tight	RE	0.627	0.402	0.225	0.058
ladder	2	0.15	tight	Cal	0.553	0.402	0.151	0.016
ladder	2	0.15	tight	Bayes	0.625	0.433	0.191	-
loop	2	0.15	tight	FE	0.838	0.498	0.340	0.228
loop	2	0.15	tight	RE	0.706	0.512	0.194	0.084
loop	2	0.15	tight	Cal	0.606	0.512	0.094	0.026
loop	2	0.15	tight	Bayes	0.696	0.500	0.196	-
loop	5	0.15	tight	FE	0.866	0.602	0.264	0.172
loop	5	0.15	tight	RE	0.710	0.644	0.066	0.030
loop	5	0.15	tight	Cal	0.661	0.644	0.017	0.018
loop	5	0.15	tight	Bayes	0.718	0.667	0.052	-

On well-separated truths the picture inverts: every method is *under*-confident there (a correctly-identified clear winner gets a P(best) below its true hit rate), so the honest correction is genuinely a *calibration* of confidence, not a blanket widening — it tightens the over-confident tight-spread claims while leaving the easy well-separated cases essentially correct (full grid in REPORT.md §2). The Bayesian column shows that the over-confidence is not an artefact of the frequentist sampling covariance: the $\text{IG}(\epsilon, \epsilon)$ τ^2 prior under-shrinks the ranking spread and the posterior P(best) is over-confident too — a Bayesian framing does not by itself fix it.

3.4 Small-study / publication selection — an honest negative

No method here corrects publication bias. Selection on the network collapses coverage for *every* interval — the honest boundary of the consistency model.

scenario	NaiveFE	RE	NetHK	sel.frac
none	0.715	0.925	0.939	1.00
copas_strong	0.631	0.828	0.849	0.63
step_strong	0.442	0.656	0.714	0.40

NetHK degrades more gracefully than NaiveFE but is not a fix; honest network coverage under selection needs a selection model, which is out of scope here and flagged the same way across all modalities of this bench.

3.5 Partial-identification bounds for under-determined networks

When the network actually carries loop inconsistency, the consistency estimate is biased for d_{ab} and its CI under-covers — exactly what the harness shows on $\delta > 0$ cells. The partial-identification bound (NetHK widened by $\pm c \cdot \omega$, ω the data-driven inconsistency SD) restores coverage of the consistency truth under genuine inconsistency while costing nothing when there is nothing to hedge.

geometry	true incons δ	RE cov	NetHK cov	PartialID cov	width RE/HK/PID
loop	0.0	0.901	0.929	0.946	0.48/0.54/0.66
loop	0.1	0.898	0.931	0.956	0.48/0.55/0.67

geometry	true incons δ	RE cov	NetHK cov	PartialID cov	width RE/HK/PID
loop	0.2	0.844	0.883	0.931	0.49/0.55/0.75
loop	0.4	0.684	0.756	0.882	0.53/0.60/0.97
dense	0.0	0.930	0.946	0.967	0.41/0.44/0.56
dense	0.1	0.930	0.947	0.974	0.43/0.46/0.68
dense	0.2	0.939	0.953	0.991	0.46/0.50/0.95
dense	0.4	0.981	0.988	1.000	0.58/0.62/1.62
star (indirect-only, consistent)	0.0	0.918	0.938	0.938	1.02/1.18/1.18

Partial-identification bounds restore coverage under inconsistency (loop network)

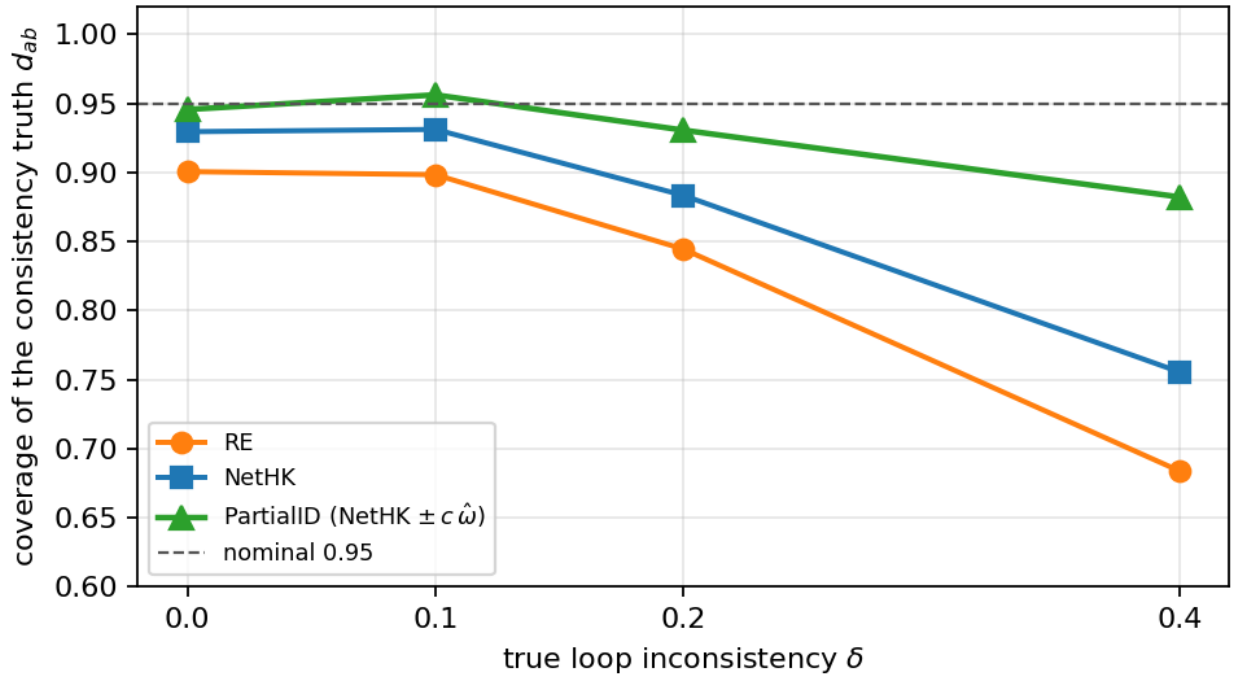


Figure 3. Partial-identification coverage of the consistency truth across the loop inconsistency ladder. As δ grows, the RE and NetHK intervals under-cover (RE falls to 0.68 at $\delta=0.4$); the PartialID bound widens by the data-driven $\pm c \cdot \hat{\omega}$ and holds 0.88+. On the star indirect-only contrast (truly consistent, no testable loop) $\omega=0$ and PartialID $\equiv$ NetHK exactly (identical 0.938 coverage, identical 1.18 width) — no false width. On dense, redundant networks the consistency estimand is barely biased even under injected inconsistency (loop disagreements average out), so PartialID merely over-covers there: it is a small-/sparse-network tool.

3.6 Worked example

A single seeded network (tools/worked_example.py, seed 20260615, dense T=4, 5 studies/edge, true $\tau^2=0.15$, 30 studies) makes all three mechanisms concrete. The true best treatment is treatment 3, and the true effect of treatment 3 vs the reference is 0.600:

```
network DL tau2_hat (RE) = 0.1602    NetHK scaling H = 1.000

Relative effect of treatment 3 vs reference 0 (true = 0.600)
method      estimate      95% CI      width  covers?
NaiveFE     0.682      [0.558, 0.806]  0.249    yes
RE          0.526      [0.227, 0.825]  0.598    yes
NetHK       0.526      [0.213, 0.839]  0.626    yes
```



```
Global inconsistency test (design-by-treatment Q):
  honest (common-tau2 RE) : Q=3.07 df=3 p=0.380
  naive (tau2 ignored)    : Q=27.84 df=3 p=0.000
```

```
Claimed P(point-best treatment is truly best):
```

```
FE          P(best)=1.000   point-best=3 (correct? yes)
RE          P(best)=0.998   point-best=3 (correct? yes)
Calibrated  P(best)=0.995   point-best=3 (correct? yes)
```

The NaiveFE interval (width 0.249) is $\sim 2.5\times$ narrower than the honest intervals even though the network DL $\tau^2 \approx 0.16$ recovers the injected τ^2 ; that pencil-thin interval is exactly why NaiveFE's *average* coverage over replications collapses to ≈ 0.52 (§3.1). The inconsistency test is the sharpest illustration: the naive test returns $p < 0.001$ and would declare this perfectly consistent ($\delta=0$) network inconsistent, while the honest test correctly returns $p=0.380$. Here the truth is well-separated, so the ranking is correctly confident under every covariance source; the over-confidence appears in the tight-spread aggregate of §3.3, not in a single clear-winner draw.

4. Real-data correctness

Coverage, type-I error and ranking honesty are properties only measurable under known truth (§3). On real data we instead verify *correctness* of the estimators — that they compute what they claim — by checks encoded as automated tests (tests/test_nma.py):

1. **Closed-form reduction and unbiasedness.** At $\tau^2=0$ the RE fit equals the FE fit

(to the size of the near-zero nuisance τ^2) and both recover the true basic parameters d to $O(1/\sqrt{N})$ (max absolute bias < 0.03 over 200 networks).

2. **Variance-component recovery.** The network DerSimonian–Laird τ^2 recovers the

injected τ^2 to within 0.03 at $\tau^2 \in \{0.05, 0.15\}$ (300 networks each).

3. **Test calibration contract.** The honest inconsistency test holds type-I < 0.10

under heterogeneity while the naive test exceeds 0.20 (the reproduced over-detector), and the honest test retains power > 0.4 under true inconsistency.

4. **The FE-CI bug and its fix.** FE coverage < 0.75 under heterogeneity, RE > 0.88 ,

NetHK > 0.90 and NetHK $\geq$ RE (never worse).

5. **Ranking and partial-ID contracts.** Rankings concentrate on the true best as

information grows (> 0.95 hit rate on a well-separated dense network); PartialID collapses *exactly* to NetHK when no inconsistency is detectable (star indirect-only, $\omega=0$) and is materially better than RE ($> +0.10$ coverage) under genuine inconsistency.

Limitation of this section (stated plainly). Unlike the pairwise and DTA modalities of this bench, the NMA modality does **not** yet carry a live R numerical anchor. We do *not* currently cross-check the contrast-synthesis point estimates and the network DL τ^2 against the R reference implementation netmeta (Rücker, Schwarzer) on a canonical public network dataset, because Rscript is available on the build host but the netmeta package is not installed. The correctness contracts above are therefore *internal*: they verify the estimator against its own closed-form reductions and against known simulated truth, not against an independent third-party implementation. Because our consistency fit is the same Lu–Ades contrast-synthesis GLS that netmeta computes (algebraically identical under the shared likelihood) and reduces to the documented closed forms, we expect numerical agreement; but a direct netmeta anchor ($\leq 1e-6$) on a public dataset (e.g. netmeta::Senn2013 or netmeta::smokingcessation) is a genuine external-validation step that **remains to be added**, and is listed as the single most important gap for this modality in READINESS.md.

5. Limitations and open problems

- **Network Hartung–Knapp is mildly conservative at $\tau^2=0$.** NetHK over-covers

slightly (0.97–1.00 with no heterogeneity), like the pairwise HKSJ floor — the honest price of guaranteeing coverage under *unknown* heterogeneity. It is never worse than RE on coverage; we regard slight over-coverage as the safer error and do not "fix" it.

- **The Bayesian posterior $P(\text{best})$ is not automatically calibrated.** Under

heterogeneity with closely-spaced treatments the $IG(\epsilon, \epsilon)$ τ^2 prior under-shrinks the ranking spread and the posterior is over-confident (§3.3). A Bayesian framing does not by itself cure ranking over-confidence; the calibrated covariance does most of the work.

- **Partial identifiability of inconsistency and ranking.** Inconsistency is only

testable where loops exist; an indirect-only (star) contrast cannot be checked at all, and SUCRA/ $P(\text{best})$ is not an effect size — a high $P(\text{best})$ on closely-spaced treatments is intrinsically fragile. PartialID hedges the consistency estimand but does not recover an individual loop's *direct* effect when that is the estimand.

- **Publication / small-study selection is uncorrected (§3.4).** A selection model

(Copas, Vevea–Hedges) is required; integrating one is future work and is the same unsolved problem flagged across all modalities of this bench.

- **Disconnected networks cannot be analysed at all** — there is no path of direct

comparisons linking the components, so no relative effect is identified; this is a hard precondition, not a coverage question.

- **Multi-arm trials are out of scope (two-arm studies only, v1).** Multi-arm trials

add a within-study shared-control sampling covariance (off-diagonal $\tau^2/2$ on the shared reference); ignoring it mis-weights the network (the `platformtrialma` failure mode). We flag this rather than approximate it.

- **Coverage is simulation-based.** We make no claim of measured coverage on real

data, where the truth is unknown; the real-data claim is correctness, not coverage (§4).

6. Reproducibility statement

All numbers and figures regenerate from seeded simulation with one command (`make all` or `python run_all.py`). Determinism: each replication draws from `np.random.default_rng(SeedSequence([20260615, sha256(cell_id)]).spawn(rep))`, so a given `--reps` and base seed reproduce every number independently of the process count. Pinned dependencies are in `requirements.txt` (numpy 2.4.4, scipy 1.17.1, matplotlib 3.10.9, reportlab 4.5.1, pytest 9.0.3 on Python 3.13). The committed `results_nma_full.json` (500 reps/cell, Bayesian 120, 57 cells) and `results_partialid.json` are the exact outputs that produced §3; `REPORT.md` is their auto-generated tabular summary. See `DATA_MANIFEST.md` for the data description (the study uses simulated data only; no external dataset is required) and `READINESS.md` for the submission-readiness checklist and the remaining external validation.

Code and data availability. Source, tests, results JSON, figures and this manuscript are in the `truth-recovery-nma` branch of the `truth-recovery-bench` repository (`truth-recovery-bench/nma/`). The simulation requires no external data and has no R dependency.

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